

3-D energy eigenvalue problem for a 'particle' in potential $V(\vec{r})$

3-D energy eigenvalue problem for a 'particle' in potential $V(\vec{r})$

'particle' in quotes : multi-particle problem may be reduced to dynamics on a single configuration coordinate with 3 components

3-D energy eigenvalue problem for a 'particle' in potential $V(\vec{r})$

'particle' in quotes : multi-particle problem may be reduced to dynamics on a single configuration coordinate with 3 components
central potential: $V = V(\vec{r}) \equiv V(r)$.

3-D energy eigenvalue problem for a 'particle' in potential $V(\vec{r})$

'particle' in quotes : multi-particle problem may be reduced to dynamics on a single configuration coordinate with 3 components

central potential: $V = V(\vec{r}) \equiv V(r)$.

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r) = -\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \right] + \frac{\hat{L}^2}{2\mu r^2} + V(r)$$

3-D energy eigenvalue problem for a 'particle' in potential $V(\vec{r})$

'particle' in quotes : multi-particle problem may be reduced to dynamics on a single configuration coordinate with 3 components

central potential: $V = V(\vec{r}) \equiv V(r)$.

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r) = -\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \right] + \frac{\hat{L}^2}{2\mu r^2} + V(r)$$

$\hat{L}^2$ has only θ, ϕ dependent terms,

3-D energy eigenvalue problem for a 'particle' in potential $V(\vec{r})$

'particle' in quotes : multi-particle problem may be reduced to dynamics on a single configuration coordinate with 3 components

central potential: $V = V(\vec{r}) \equiv V(r)$.

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r) = -\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \right] + \frac{\hat{L}^2}{2\mu r^2} + V(r)$$

$\hat{L}^2$ has only θ, ϕ dependent terms, $\therefore [\hat{L}^2, \hat{H}] = 0$; and $[\hat{L}_z, \hat{H}] = 0$

3-D energy eigenvalue problem for a 'particle' in potential $V(\vec{r})$

'particle' in quotes : multi-particle problem may be reduced to dynamics on a single configuration coordinate with 3 components

central potential: $V = V(\vec{r}) \equiv V(r)$.

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r) = -\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \right] + \frac{\hat{L}^2}{2\mu r^2} + V(r)$$

$\hat{L}^2$ has only θ, ϕ dependent terms, $\therefore [\hat{L}^2, \hat{H}] = 0$; and $[\hat{L}_z, \hat{H}] = 0$

Also, $[\hat{L}^2, \hat{L}_z] = 0$;

3-D energy eigenvalue problem for a 'particle' in potential $V(\vec{r})$

'particle' in quotes : multi-particle problem may be reduced to dynamics on a single configuration coordinate with 3 components

central potential: $V = V(\vec{r}) \equiv V(r)$.

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r) = -\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \right] + \frac{\hat{L}^2}{2\mu r^2} + V(r)$$

$\hat{L}^2$ has only θ, ϕ dependent terms, $\therefore [\hat{L}^2, \hat{H}] = 0$; and $[\hat{L}_z, \hat{H}] = 0$

Also, $[\hat{L}^2, \hat{L}_z] = 0$; $\therefore$ simultaneous eigenfunctions of $\hat{L}^2$, $\hat{L}_z$, and $\hat{H}$.

3-D energy eigenvalue problem for a 'particle' in potential $V(\vec{r})$

'particle' in quotes : multi-particle problem may be reduced to dynamics on a single configuration coordinate with 3 components

central potential: $V = V(\vec{r}) \equiv V(r)$.

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r) = -\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \right] + \frac{\hat{L}^2}{2\mu r^2} + V(r)$$

$\hat{L}^2$ has only θ, ϕ dependent terms, $\therefore [\hat{L}^2, \hat{H}] = 0$; and $[\hat{L}_z, \hat{H}] = 0$

Also, $[\hat{L}^2, \hat{L}_z] = 0$; $\therefore$ simultaneous eigenfunctions of $\hat{L}^2$, $\hat{L}_z$, and $\hat{H}$.

solutions for the angular part : spherical harmonics

3-D energy eigenvalue problem for a 'particle' in potential $V(\vec{r})$

'particle' in quotes : multi-particle problem may be reduced to dynamics on a single configuration coordinate with 3 components

central potential: $V = V(\vec{r}) \equiv V(r)$.

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r) = -\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \right] + \frac{\hat{L}^2}{2\mu r^2} + V(r)$$

$\hat{L}^2$ has only θ, ϕ dependent terms, $\therefore [\hat{L}^2, \hat{H}] = 0$; and $[\hat{L}_z, \hat{H}] = 0$

Also, $[\hat{L}^2, \hat{L}_z] = 0$; $\therefore$ simultaneous eigenfunctions of $\hat{L}^2$, $\hat{L}_z$, and $\hat{H}$.

solutions for the angular part : spherical harmonics

complete solution : $\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$; $\Psi(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}) e^{-\frac{iE_n t}{\hbar}}$

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] \psi + \frac{\hat{L}^2}{2\mu r^2} \psi + V\psi = E\psi \text{ [note : } \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} r]$$

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] \psi + \frac{\hat{L}^2}{2\mu r^2} \psi + V\psi = E\psi \text{ [note : } \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} r]$$

$$\hat{L}^2 Y_l^m(\Omega) = l(l+1)\hbar^2 Y_l^m(\Omega)$$

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] \psi + \frac{\hat{L}^2}{2\mu r^2} \psi + V\psi = E\psi \quad [\text{note : } \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} r]$$

$$\hat{L}^2 Y_l^m(\Omega) = l(l+1)\hbar^2 Y_l^m(\Omega)$$

$$\therefore -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) Y_l^m(\Omega) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) Y_l^m(\Omega) + V \cdot R(r) Y_l^m(\Omega) = E R(r) Y_l^m(\Omega)$$

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] \psi + \frac{\hat{L}^2}{2\mu r^2} \psi + V\psi = E\psi \quad [\text{note : } \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} r]$$

$$\hat{L}^2 Y_l^m(\Omega) = l(l+1)\hbar^2 Y_l^m(\Omega)$$

$$\therefore -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) Y_l^m(\Omega) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) Y_l^m(\Omega) + V \cdot R(r) Y_l^m(\Omega) = E R(r) Y_l^m(\Omega)$$

$$\text{divide by } Y_l^m(\Omega), \quad -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) + V \cdot R(r) = E R(r)$$

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] \psi + \frac{\hat{L}^2}{2\mu r^2} \psi + V\psi = E\psi \quad [\text{note : } \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} r]$$

$$\hat{L}^2 Y_l^m(\Omega) = l(l+1)\hbar^2 Y_l^m(\Omega)$$

$$\therefore -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) Y_l^m(\Omega) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) Y_l^m(\Omega) + V \cdot R(r) Y_l^m(\Omega) = E R(r) Y_l^m(\Omega)$$

$$\text{divide by } Y_l^m(\Omega), \quad -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) + V \cdot R(r) = E R(r)$$

write $u = rR$;

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] \psi + \frac{\hat{L}^2}{2\mu r^2} \psi + V\psi = E\psi \quad [\text{note : } \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} r]$$

$$\hat{L}^2 Y_l^m(\Omega) = l(l+1)\hbar^2 Y_l^m(\Omega)$$

$$\therefore -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) Y_l^m(\Omega) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) Y_l^m(\Omega) + V \cdot R(r) Y_l^m(\Omega) = E R(r) Y_l^m(\Omega)$$

$$\text{divide by } Y_l^m(\Omega), \quad -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) + V \cdot R(r) = E R(r)$$

$$\text{write } u = rR; \text{ and get } \frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$$

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] \psi + \frac{\hat{L}^2}{2\mu r^2} \psi + V\psi = E\psi \quad [\text{note : } \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} r]$$

$$\hat{L}^2 Y_l^m(\Omega) = l(l+1)\hbar^2 Y_l^m(\Omega)$$

$$\therefore -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) Y_l^m(\Omega) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) Y_l^m(\Omega) + V \cdot R(r) Y_l^m(\Omega) = E R(r) Y_l^m(\Omega)$$

$$\text{divide by } Y_l^m(\Omega), \quad -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) + V \cdot R(r) = E R(r)$$

$$\text{write } u = rR; \text{ and get } \frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$$

This equation looks like 1-D SE with a few differences:

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] \psi + \frac{\hat{L}^2}{2\mu r^2} \psi + V\psi = E\psi \quad [\text{note} : \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} r]$$

$$\hat{L}^2 Y_l^m(\Omega) = l(l+1)\hbar^2 Y_l^m(\Omega)$$

$$\therefore -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) Y_l^m(\Omega) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) Y_l^m(\Omega) + V \cdot R(r) Y_l^m(\Omega) = E R(r) Y_l^m(\Omega)$$

$$\text{divide by } Y_l^m(\Omega), \quad -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) + V \cdot R(r) = E R(r)$$

$$\text{write } u = rR; \text{ and get } \frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$$

This equation looks like 1-D SE with a few differences:

(i) extra term for potential $\frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2}$: centrifugal contribution

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] \psi + \frac{\hat{L}^2}{2\mu r^2} \psi + V\psi = E\psi \quad [\text{note : } \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} r]$$

$$\hat{L}^2 Y_l^m(\Omega) = l(l+1)\hbar^2 Y_l^m(\Omega)$$

$$\therefore -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) Y_l^m(\Omega) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) Y_l^m(\Omega) + V.R(r) Y_l^m(\Omega) = ER(r) Y_l^m(\Omega)$$

$$\text{divide by } Y_l^m(\Omega), -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) + V.R(r) = ER(r)$$

$$\text{write } u = rR; \text{ and get } \frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$$

This equation looks like 1-D SE with a few differences:

- (i) extra term for potential $\frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2}$: centrifugal contribution
- (ii) u must satisfy $u(0) = 0$, so that ψ is well behaved;

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\Omega)$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] \psi + \frac{\hat{L}^2}{2\mu r^2} \psi + V\psi = E\psi \quad [\text{note : } \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial^2}{\partial r^2} r]$$

$$\hat{L}^2 Y_l^m(\Omega) = l(l+1)\hbar^2 Y_l^m(\Omega)$$

$$\therefore -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) Y_l^m(\Omega) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) Y_l^m(\Omega) + V.R(r) Y_l^m(\Omega) = ER(r) Y_l^m(\Omega)$$

$$\text{divide by } Y_l^m(\Omega), \quad -\frac{\hbar^2}{2\mu} \left[\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right] R(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} R(r) + V.R(r) = ER(r)$$

$$\text{write } u = rR; \text{ and get } \frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$$

This equation looks like 1-D SE with a few differences:

- (i) extra term for potential $\frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2}$: centrifugal contribution
- (ii) u must satisfy $u(0) = 0$, so that ψ is well behaved; (iii) $0 < r < \infty$

condition $u(0) = 0$ makes the solution unique

condition $u(0) = 0$ makes the solution unique

If there existed two solutions, u_1 and u_2 , then $u_1(0) = u_2(0) = 0$

condition $u(0) = 0$ makes the solution unique

If there existed two solutions, u_1 and u_2 , then $u_1(0) = u_2(0) = 0$

$$u_2 \times \left\{ \frac{d^2 u_1}{dr^2} + \frac{2\mu E}{\hbar^2} u_1 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_1 \right.$$

condition $u(0) = 0$ makes the solution unique

If there existed two solutions, u_1 and u_2 , then $u_1(0) = u_2(0) = 0$

$$\begin{aligned} u_2 \times \left\{ \frac{d^2 u_1}{dr^2} + \frac{2\mu E}{\hbar^2} u_1 \right. &= \left. \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_1 \right. \\ u_1 \times \left\{ \frac{d^2 u_2}{dr^2} + \frac{2\mu E}{\hbar^2} u_2 \right. &= \left. \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_2 \right. \end{aligned}$$

condition $u(0) = 0$ makes the solution unique

If there existed two solutions, u_1 and u_2 , then $u_1(0) = u_2(0) = 0$

$$u_2 \times \left\{ \frac{d^2 u_1}{dr^2} + \frac{2\mu E}{\hbar^2} u_1 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_1 \right.$$

$$\left. u_1 \times \left\{ \frac{d^2 u_2}{dr^2} + \frac{2\mu E}{\hbar^2} u_2 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_2 \right. \right.$$

$$\text{subtract : } u_2 u_1'' - u_1 u_2'' = d(u_2 u_1' - u_1 u_2') = 0$$

condition $u(0) = 0$ makes the solution unique

If there existed two solutions, u_1 and u_2 , then $u_1(0) = u_2(0) = 0$

$$u_2 \times \left\{ \frac{d^2 u_1}{dr^2} + \frac{2\mu E}{\hbar^2} u_1 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_1 \right.$$

$$\left. u_1 \times \left\{ \frac{d^2 u_2}{dr^2} + \frac{2\mu E}{\hbar^2} u_2 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_2 \right. \right.$$

$$\text{subtract : } u_2 u_1'' - u_1 u_2'' = d(u_2 u_1' - u_1 u_2') = 0$$

$$\therefore u_2 u_1' - u_1 u_2' = \text{constant};$$

condition $u(0) = 0$ makes the solution unique

If there existed two solutions, u_1 and u_2 , then $u_1(0) = u_2(0) = 0$

$$u_2 \times \left\{ \frac{d^2 u_1}{dr^2} + \frac{2\mu E}{\hbar^2} u_1 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_1 \right.$$

$$u_1 \times \left\{ \frac{d^2 u_2}{dr^2} + \frac{2\mu E}{\hbar^2} u_2 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_2 \right.$$

$$\text{subtract : } u_2 u_1'' - u_1 u_2'' = d(u_2 u_1' - u_1 u_2') = 0$$

$$\therefore u_2 u_1' - u_1 u_2' = \text{constant}; \text{ using B.C. } u_1(0) = u_2(0) = 0,$$

condition $u(0) = 0$ makes the solution unique

If there existed two solutions, u_1 and u_2 , then $u_1(0) = u_2(0) = 0$

$$u_2 \times \left\{ \frac{d^2 u_1}{dr^2} + \frac{2\mu E}{\hbar^2} u_1 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_1 \right.$$

$$u_1 \times \left\{ \frac{d^2 u_2}{dr^2} + \frac{2\mu E}{\hbar^2} u_2 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_2 \right.$$

$$\text{subtract : } u_2 u_1'' - u_1 u_2'' = d(u_2 u_1' - u_1 u_2') = 0$$

$$\therefore u_2 u_1' - u_1 u_2' = \text{constant}; \text{ using B.C. } u_1(0) = u_2(0) = 0, \text{ constant} = \text{zero}.$$

condition $u(0) = 0$ makes the solution unique

If there existed two solutions, u_1 and u_2 , then $u_1(0) = u_2(0) = 0$

$$u_2 \times \left\{ \frac{d^2 u_1}{dr^2} + \frac{2\mu E}{\hbar^2} u_1 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_1 \right.$$

$$u_1 \times \left\{ \frac{d^2 u_2}{dr^2} + \frac{2\mu E}{\hbar^2} u_2 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_2 \right.$$

$$\text{subtract : } u_2 u_1'' - u_1 u_2'' = d(u_2 u_1' - u_1 u_2') = 0$$

$\therefore u_2 u_1' - u_1 u_2' = \text{constant}$; using B.C. $u_1(0) = u_2(0) = 0$, constant = zero.

$$\therefore u_2 u_1' = u_1 u_2' \implies \frac{u_2'}{u_2} = \frac{u_1'}{u_1} \implies u_2 = a u_1, \text{ where } a \text{ is constant.}$$

condition $u(0) = 0$ makes the solution unique

If there existed two solutions, u_1 and u_2 , then $u_1(0) = u_2(0) = 0$

$$u_2 \times \left\{ \frac{d^2 u_1}{dr^2} + \frac{2\mu E}{\hbar^2} u_1 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_1 \right.$$

$$u_1 \times \left\{ \frac{d^2 u_2}{dr^2} + \frac{2\mu E}{\hbar^2} u_2 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_2 \right.$$

$$\text{subtract : } u_2 u_1'' - u_1 u_2'' = d(u_2 u_1' - u_1 u_2') = 0$$

$\therefore u_2 u_1' - u_1 u_2' = \text{constant}$; using B.C. $u_1(0) = u_2(0) = 0$, constant = zero.

$$\therefore u_2 u_1' = u_1 u_2' \implies \frac{u_2'}{u_2} = \frac{u_1'}{u_1} \implies u_2 = a u_1, \text{ where } a \text{ is constant.}$$

$\therefore$ for given E , no two linearly independent solutions can satisfy $f(0) = 0$ simultaneously

condition $u(0) = 0$ makes the solution unique

If there existed two solutions, u_1 and u_2 , then $u_1(0) = u_2(0) = 0$

$$u_2 \times \left\{ \frac{d^2 u_1}{dr^2} + \frac{2\mu E}{\hbar^2} u_1 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_1 \right.$$

$$u_1 \times \left\{ \frac{d^2 u_2}{dr^2} + \frac{2\mu E}{\hbar^2} u_2 = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u_2 \right.$$

$$\text{subtract : } u_2 u_1'' - u_1 u_2'' = d(u_2 u_1' - u_1 u_2') = 0$$

$\therefore u_2 u_1' - u_1 u_2' = \text{constant}$; using B.C. $u_1(0) = u_2(0) = 0$, constant = zero.

$$\therefore u_2 u_1' = u_1 u_2' \implies \frac{u_2'}{u_2} = \frac{u_1'}{u_1} \implies u_2 = a u_1, \text{ where } a \text{ is constant.}$$

$\therefore$ for given E , no two linearly independent solutions can satisfy $f(0) = 0$ simultaneously

$$\therefore \psi(r, \theta, \phi) = \frac{u(r)}{r} Y_l^m(\Omega)$$

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

Cartesian Coordinates, solution : $\frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\vec{k} \cdot \vec{r}}$

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

Cartesian Coordinates, solution : $\frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\vec{k} \cdot \vec{r}}$

$\vec{k}$ oriented in any direction : infinite degeneracy

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

Cartesian Coordinates, solution : $\frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\vec{k} \cdot \vec{r}}$

$\vec{k}$ oriented in any direction : infinite degeneracy
states of definite energy and angular momentum

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

Cartesian Coordinates, solution : $\frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\vec{k} \cdot \vec{r}}$

$\vec{k}$ oriented in any direction : infinite degeneracy
states of definite energy and angular momentum
no quantization in energy;

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

Cartesian Coordinates, solution : $\frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\vec{k} \cdot \vec{r}}$

$\vec{k}$ oriented in any direction : infinite degeneracy
states of definite energy and angular momentum
no quantization in energy;

Let $\rho = kr$, $u = \sqrt{\rho} v$ and $\nu = (l + \frac{1}{2})$

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

Cartesian Coordinates, solution : $\frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\vec{k} \cdot \vec{r}}$

$\vec{k}$ oriented in any direction : infinite degeneracy
states of definite energy and angular momentum
no quantization in energy;

Let $\rho = kr$, $u = \sqrt{\rho} v$ and $\nu = (l + \frac{1}{2})$

SE : Bessel's equation

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

Cartesian Coordinates, solution : $\frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\vec{k} \cdot \vec{r}}$

$\vec{k}$ oriented in any direction : infinite degeneracy
states of definite energy and angular momentum
no quantization in energy;

Let $\rho = kr$, $u = \sqrt{\rho} v$ and $\nu = (l + \frac{1}{2})$

SE : Bessel's equation $\rho^2 \frac{d^2 v}{d\rho^2} + \rho \frac{dv}{d\rho} + [\rho^2 - \nu^2] v = 0$

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

Cartesian Coordinates, solution : $\frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\vec{k} \cdot \vec{r}}$

$\vec{k}$ oriented in any direction : infinite degeneracy
states of definite energy and angular momentum
no quantization in energy;

Let $\rho = kr$, $u = \sqrt{\rho} v$ and $\nu = (l + \frac{1}{2})$

SE : Bessel's equation $\rho^2 \frac{d^2 v}{d\rho^2} + \rho \frac{dv}{d\rho} + [\rho^2 - \nu^2] v = 0$
two independent solutions, $v \Rightarrow J_{\pm\nu}(\rho)$

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

Cartesian Coordinates, solution : $\frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\vec{k} \cdot \vec{r}}$

$\vec{k}$ oriented in any direction : infinite degeneracy
states of definite energy and angular momentum
no quantization in energy;

Let $\rho = kr$, $u = \sqrt{\rho} v$ and $\nu = (l + \frac{1}{2})$

SE : Bessel's equation $\rho^2 \frac{d^2 v}{d\rho^2} + \rho \frac{dv}{d\rho} + [\rho^2 - \nu^2] v = 0$

two independent solutions, $v \Rightarrow J_{\pm\nu}(\rho)$

complete solution ($R = \frac{u}{r} = \frac{\sqrt{\rho} v}{r} = \frac{k}{\sqrt{\rho}} v$) $\Rightarrow \psi(r, \theta, \phi) = \frac{k J_{\nu}(\rho)}{\sqrt{\rho}} Y_l^m(\Omega)$

free particle: $V(r) = 0$

$$E_k = \frac{\hbar^2 k^2}{2\mu}; \quad k^2 = \frac{2\mu E}{\hbar^2}; \quad \frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$$

Cartesian Coordinates, solution : $\frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\vec{k} \cdot \vec{r}}$

$\vec{k}$ oriented in any direction : infinite degeneracy
states of definite energy and angular momentum
no quantization in energy;

Let $\rho = kr$, $u = \sqrt{\rho} v$ and $\nu = (l + \frac{1}{2})$

SE : Bessel's equation $\rho^2 \frac{d^2 v}{d\rho^2} + \rho \frac{dv}{d\rho} + [\rho^2 - \nu^2] v = 0$

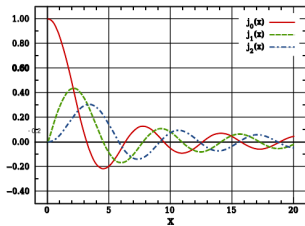
two independent solutions, $v \Rightarrow J_{\pm\nu}(\rho)$

complete solution ($R = \frac{u}{r} = \frac{\sqrt{\rho} v}{r} = \frac{k}{\sqrt{\rho}} v$) $\Rightarrow \psi(r, \theta, \phi) = \frac{k J_{\nu}(\rho)}{\sqrt{\rho}} Y_l^m(\Omega)$

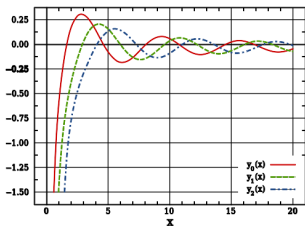
expressed as spherical Bessel function

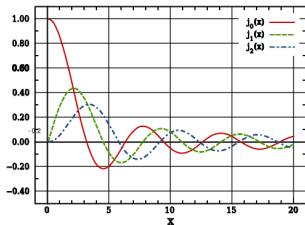
1st kind $j_l(\rho) = \sqrt{\frac{\pi}{2\rho}} J_{l+\frac{1}{2}}(\rho)$

2nd kind $\eta_l(\rho) = (-1)^{l+1} \sqrt{\frac{\pi}{2\rho}} J_{-l+\frac{1}{2}}(\rho)$

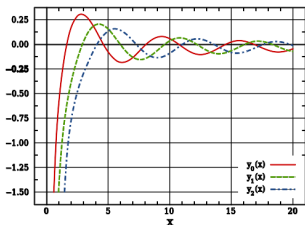


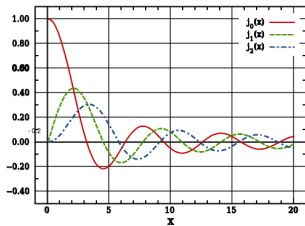
as $\rho \rightarrow 0$, $j_l = \frac{\rho^l}{1.3.5 \dots (2l+1)} \rightarrow$ gives regular soln



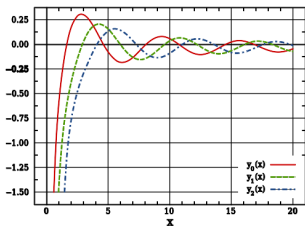


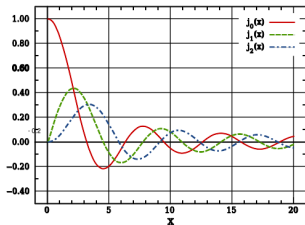
as $\rho \rightarrow 0$, $j_l = \frac{\rho^l}{1.3.5 \dots (2l+1)} \rightarrow$ gives regular soln
 and $\eta_l = -\frac{1.3 \dots (2l-1)}{\rho^{l+1}} \rightarrow \infty$ even for $l = 0$,



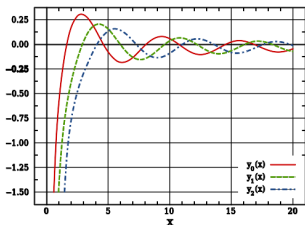


as $\rho \rightarrow 0$, $j_l = \frac{\rho^l}{1.3.5 \dots (2l+1)} \rightarrow$ gives regular soln
 and $\eta_l = -\frac{1.3 \dots (2l-1)}{\rho^{l+1}} \rightarrow \infty$ even for $l = 0$,
 not acceptable





as $\rho \rightarrow 0$, $j_l = \frac{\rho^l}{1.3.5 \dots (2l+1)} \rightarrow$ gives regular soln
 and $\eta_l = -\frac{1.3 \dots (2l-1)}{\rho^{l+1}} \rightarrow \infty$ even for $l = 0$,
 not acceptable



as $\rho \rightarrow \infty$, $j_l \rightarrow \frac{1}{\rho} \sin(\rho - \frac{l\pi}{2})$, ok $\because \frac{1}{\rho} \rightarrow 0$
 and $\sin(\rho - \frac{l\pi}{2})$ is oscillatory

Back to $\frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$

Back to $\frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$

For $l = 0$, $\frac{d^2 u_0}{dr^2} + k^2 u_0 = 0$

Back to $\frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$

For $l = 0$, $\frac{d^2 u_0}{dr^2} + k^2 u_0 = 0 \implies u_0 = \sin kr$ or $\cos kr$

Back to $\frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$

For $l = 0$, $\frac{d^2 u_0}{dr^2} + k^2 u_0 = 0 \implies u_0 = \sin kr$ or $\cos kr$

Thus R for $l = 0$ is a combination of $\frac{\sin kr}{kr}$ and $\frac{\cos kr}{kr}$

Back to $\frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$

For $l = 0$, $\frac{d^2 u_0}{dr^2} + k^2 u_0 = 0 \implies u_0 = \sin kr$ or $\cos kr$

Thus R for $l = 0$ is a combination of $\frac{\sin kr}{kr}$ and $\frac{\cos kr}{kr}$

acceptable solution: $\frac{\sin kr}{kr} \xrightarrow{r \rightarrow 0} 1$

Back to $\frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$

For $l = 0$, $\frac{d^2 u_0}{dr^2} + k^2 u_0 = 0 \implies u_0 = \sin kr$ or $\cos kr$

Thus R for $l = 0$ is a combination of $\frac{\sin kr}{kr}$ and $\frac{\cos kr}{kr}$

acceptable solution: $\frac{\sin kr}{kr} \xrightarrow{r \rightarrow 0} 1$

$\therefore$ for each $E : \psi(r, \theta, \phi) \sim j_l(kr) Y_l^m(\Omega)$

Back to $\frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$

For $l = 0$, $\frac{d^2 u_0}{dr^2} + k^2 u_0 = 0 \implies u_0 = \sin kr$ or $\cos kr$

Thus R for $l = 0$ is a combination of $\frac{\sin kr}{kr}$ and $\frac{\cos kr}{kr}$

acceptable solution: $\frac{\sin kr}{kr} \xrightarrow{r \rightarrow 0} 1$

$\therefore$ for each $E : \psi(r, \theta, \phi) \sim j_l(kr) Y_l^m(\Omega)$

Things are complex when we deal with a non-zero potential field,

Back to $\frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$

For $l = 0$, $\frac{d^2 u_0}{dr^2} + k^2 u_0 = 0 \implies u_0 = \sin kr$ or $\cos kr$

Thus R for $l = 0$ is a combination of $\frac{\sin kr}{kr}$ and $\frac{\cos kr}{kr}$

acceptable solution: $\frac{\sin kr}{kr} \xrightarrow{r \rightarrow 0} 1$

$$\therefore \text{for each } E : \psi(r, \theta, \phi) \sim j_l(kr) Y_l^m(\Omega)$$

Things are complex when we deal with a non-zero potential field,

e.g., hydrogenic atom with a nucleus of charge $+Ze$ and an electron of charge $-e$

Back to $\frac{d^2 u}{dr^2} + k^2 u = \frac{l(l+1)}{r^2} u$

For $l = 0$, $\frac{d^2 u_0}{dr^2} + k^2 u_0 = 0 \implies u_0 = \sin kr$ or $\cos kr$

Thus R for $l = 0$ is a combination of $\frac{\sin kr}{kr}$ and $\frac{\cos kr}{kr}$

acceptable solution: $\frac{\sin kr}{kr} \xrightarrow{r \rightarrow 0} 1$

$$\therefore \text{for each } E : \psi(r, \theta, \phi) \sim j_l(kr) Y_l^m(\Omega)$$

Things are complex when we deal with a non-zero potential field,

e.g., hydrogenic atom with a nucleus of charge $+Ze$ and an electron of charge $-e$

with Coulomb interaction, $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$

H atom

H atom

most dramatic success in history of QM :

understanding of details of spectra of simple atoms and that of the periodicities found in the table of chemical elements

Approximations:

H atom

most dramatic success in history of QM :
understanding of details of spectra of simple atoms and that of the periodicities found in the table of chemical elements

Approximations:

1. Proton Heavy: $\frac{m_e}{m_p} \sim \frac{1}{1836}$;

H atom

most dramatic success in history of QM :

understanding of details of spectra of simple atoms and that of the periodicities found in the table of chemical elements

Approximations:

1. Proton Heavy: $\frac{m_e}{m_p} \sim \frac{1}{1836}$;

take limit $m_p \rightarrow \infty$, corrections to this can be done later

H atom

most dramatic success in history of QM :

understanding of details of spectra of simple atoms and that of the periodicities found in the table of chemical elements

Approximations:

1. Proton Heavy: $\frac{m_e}{m_p} \sim \frac{1}{1836}$;
take limit $m_p \longrightarrow \infty$, corrections to this can be done later
2. Forget spin :

H atom

most dramatic success in history of QM :

understanding of details of spectra of simple atoms and that of the periodicities found in the table of chemical elements

Approximations:

1. Proton Heavy: $\frac{m_e}{m_p} \sim \frac{1}{1836}$;
take limit $m_p \longrightarrow \infty$, corrections to this can be done later
2. Forget spin :
use non-relativistic mechanics and do corrections later

H atom

most dramatic success in history of QM :

understanding of details of spectra of simple atoms and that of the periodicities found in the table of chemical elements

Approximations:

1. Proton Heavy: $\frac{m_e}{m_p} \sim \frac{1}{1836}$;

take limit $m_p \rightarrow \infty$, corrections to this can be done later

2. Forget spin :

use non-relativistic mechanics and do corrections later

small magnetic effects arise because from POV of electron, proton is a circulating charge which produces a magnetic field;

H atom

most dramatic success in history of QM :

understanding of details of spectra of simple atoms and that of the periodicities found in the table of chemical elements

Approximations:

1. Proton Heavy: $\frac{m_e}{m_p} \sim \frac{1}{1836}$;

take limit $m_p \rightarrow \infty$, corrections to this can be done later

2. Forget spin :

use non-relativistic mechanics and do corrections later

small magnetic effects arise because from POV of electron, proton is a circulating charge which produces a magnetic field;

in this field, the electron has a different energy with spin up than when spin is down;

H atom

most dramatic success in history of QM :

understanding of details of spectra of simple atoms and that of the periodicities found in the table of chemical elements

Approximations:

1. Proton Heavy: $\frac{m_e}{m_p} \sim \frac{1}{1836}$;

take limit $m_p \rightarrow \infty$, corrections to this can be done later

2. Forget spin :

use non-relativistic mechanics and do corrections later

small magnetic effects arise because from POV of electron, proton is a circulating charge which produces a magnetic field;

in this field, the electron has a different energy with spin up than when spin is down; ignore this small energy shift

H atom

most dramatic success in history of QM :

understanding of details of spectra of simple atoms and that of the periodicities found in the table of chemical elements

Approximations:

1. Proton Heavy: $\frac{m_e}{m_p} \sim \frac{1}{1836}$;

take limit $m_p \rightarrow \infty$, corrections to this can be done later

2. Forget spin :

use non-relativistic mechanics and do corrections later

small magnetic effects arise because from POV of electron, proton is a circulating charge which produces a magnetic field;

in this field, the electron has a different energy with spin up than when spin is down; ignore this small energy shift

3. With fixed spin, for a fixed distance of electron from nucleus, atom is a free rotational system in space $\implies$ angular momentum is conserved

H atom

most dramatic success in history of QM :

understanding of details of spectra of simple atoms and that of the periodicities found in the table of chemical elements

Approximations:

1. Proton Heavy: $\frac{m_e}{m_p} \sim \frac{1}{1836}$;

take limit $m_p \rightarrow \infty$, corrections to this can be done later

2. Forget spin :

use non-relativistic mechanics and do corrections later

small magnetic effects arise because from POV of electron, proton is a circulating charge which produces a magnetic field;

in this field, the electron has a different energy with spin up than when spin is down; ignore this small energy shift

3. With fixed spin, for a fixed distance of electron from nucleus, atom is a free rotational system in space $\Rightarrow$ angular momentum is conserved
transform two-body problem to C.M. and relative coordinate framework

states of definite ang momentum with quantum nos. l , m

states of definite ang momentum with quantum nos. l, m : $\psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

states of definite ang momentum with quantum nos. l, m : $\psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

states of definite ang momentum with quantum nos. l, m : $\psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

consider free (ionized) electron to be of zero energy (reference).

states of definite angular momentum with quantum nos. l, m : $\psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

consider free (ionized) electron to be of zero energy (reference).

bound state problem, $E < 0$; $E = -\varepsilon$ where $\varepsilon > 0$

states of definite angular momentum with quantum nos. l, m : $\psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

consider free (ionized) electron to be of zero energy (reference).

bound state problem, $E < 0$; $E = -\varepsilon$ where $\varepsilon > 0$

using $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$, SE: $\frac{d^2 u}{dr^2} - \frac{2\mu\varepsilon}{\hbar^2} u = \left[-\frac{2\mu}{\hbar^2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{l(l+1)}{r^2} \right] u$ with $u(0) = 0$

states of definite ang momentum with quantum nos. l, m : $\psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

consider free (ionized) electron to be of zero energy (reference).

bound state problem, $E < 0$; $E = -\varepsilon$ where $\varepsilon > 0$

using $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$, SE: $\frac{d^2 u}{dr^2} - \frac{2\mu\varepsilon}{\hbar^2} u = \left[-\frac{2\mu}{\hbar^2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{l(l+1)}{r^2} \right] u$ with $u(0) = 0$

Define:

$$\kappa = \sqrt{\frac{2\mu\varepsilon}{\hbar^2}}; \quad \rho_0 = \frac{\mu Ze^2}{2\pi\epsilon_0 \kappa \hbar^2} = \frac{Ze^2}{2\pi\epsilon_0 \hbar} \sqrt{\frac{\mu}{2\varepsilon}}; \text{ and let } \rho = \kappa r (\text{dimensionless})$$

states of definite ang momentum with quantum nos. l, m : $\psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

consider free (ionized) electron to be of zero energy (reference).

bound state problem, $E < 0$; $E = -\varepsilon$ where $\varepsilon > 0$

using $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$, SE: $\frac{d^2 u}{dr^2} - \frac{2\mu\varepsilon}{\hbar^2} u = \left[-\frac{2\mu}{\hbar^2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{l(l+1)}{r^2} \right] u$ with $u(0) = 0$

Define:

$$\kappa = \sqrt{\frac{2\mu\varepsilon}{\hbar^2}}; \quad \rho_0 = \frac{\mu Ze^2}{2\pi\epsilon_0 \kappa \hbar^2} = \frac{Ze^2}{2\pi\epsilon_0 \hbar} \sqrt{\frac{\mu}{2\varepsilon}}; \text{ and let } \rho = \kappa r (\text{dimensionless})$$

$$\frac{d^2 u}{d\rho^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2} \right] u, \quad \text{with } u(0) = 0$$

states of definite ang momentum with quantum nos. l, m : $\psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

consider free (ionized) electron to be of zero energy (reference).

bound state problem, $E < 0$; $E = -\varepsilon$ where $\varepsilon > 0$

using $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$, SE: $\frac{d^2 u}{dr^2} - \frac{2\mu\varepsilon}{\hbar^2} u = \left[-\frac{2\mu}{\hbar^2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{l(l+1)}{r^2} \right] u$ with $u(0) = 0$

Define:

$$\kappa = \sqrt{\frac{2\mu\varepsilon}{\hbar^2}}; \quad \rho_0 = \frac{\mu Ze^2}{2\pi\epsilon_0 \kappa \hbar^2} = \frac{Ze^2}{2\pi\epsilon_0 \hbar} \sqrt{\frac{\mu}{2\varepsilon}}; \text{ and let } \rho = \kappa r (\text{dimensionless})$$

$$\frac{d^2 u}{d\rho^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2} \right] u, \quad \text{with } u(0) = 0$$

$$\begin{array}{l} \rho \rightarrow 0, \quad \frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u = 0 \\ \implies u \propto \rho^{l+1} (\text{good}) \text{ or } \rho^{-l} (\text{bad}) \end{array} \quad \left| \right.$$

states of definite ang momentum with quantum nos. l, m : $\psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

consider free (ionized) electron to be of zero energy (reference).

bound state problem, $E < 0$; $E = -\varepsilon$ where $\varepsilon > 0$

using $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$, SE: $\frac{d^2 u}{dr^2} - \frac{2\mu\varepsilon}{\hbar^2} u = \left[-\frac{2\mu}{\hbar^2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{l(l+1)}{r^2} \right] u$ with $u(0) = 0$

Define:

$$\kappa = \sqrt{\frac{2\mu\varepsilon}{\hbar^2}}; \quad \rho_0 = \frac{\mu Ze^2}{2\pi\epsilon_0 \kappa \hbar^2} = \frac{Ze^2}{2\pi\epsilon_0 \hbar} \sqrt{\frac{\mu}{2\varepsilon}}; \text{ and let } \rho = \kappa r (\text{dimensionless})$$

$$\frac{d^2 u}{d\rho^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2} \right] u, \quad \text{with } u(0) = 0$$

$$\begin{array}{l|l} \rho \rightarrow 0, \quad \frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u = 0 & \rho \rightarrow \infty, \quad \frac{d^2 u}{d\rho^2} = u \\ \implies u \propto \rho^{l+1} (\text{good}) \text{ or } \rho^{-l} (\text{bad}) & \implies u \propto e^{-\rho} (\text{good}) \text{ or } e^{\rho} (\text{bad}) \end{array}$$

states of definite ang momentum with quantum nos. $l, m : \psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

consider free (ionized) electron to be of zero energy (reference).

bound state problem, $E < 0$; $E = -\varepsilon$ where $\varepsilon > 0$

using $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$, SE: $\frac{d^2 u}{dr^2} - \frac{2\mu\varepsilon}{\hbar^2} u = \left[-\frac{2\mu}{\hbar^2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{l(l+1)}{r^2} \right] u$ with $u(0) = 0$

Define:

$$\kappa = \sqrt{\frac{2\mu\varepsilon}{\hbar^2}}; \quad \rho_0 = \frac{\mu Ze^2}{2\pi\epsilon_0 \kappa \hbar^2} = \frac{Ze^2}{2\pi\epsilon_0 \hbar} \sqrt{\frac{\mu}{2\varepsilon}}; \text{ and let } \rho = \kappa r (\text{dimensionless})$$

$$\frac{d^2 u}{d\rho^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2} \right] u, \quad \text{with } u(0) = 0$$

$$\begin{array}{l|l} \rho \rightarrow 0, \quad \frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u = 0 & \rho \rightarrow \infty, \quad \frac{d^2 u}{d\rho^2} = u \\ \implies u \propto \rho^{l+1} (\text{good}) \text{ or } \rho^{-l} (\text{bad}) & \implies u \propto e^{-\rho} (\text{good}) \text{ or } e^{\rho} (\text{bad}) \end{array}$$
$$\therefore u(\rho) = \rho^{l+1} e^{-\rho} v(\rho)$$

states of definite ang momentum with quantum nos. $l, m : \psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

consider free (ionized) electron to be of zero energy (reference).

bound state problem, $E < 0$; $E = -\varepsilon$ where $\varepsilon > 0$

using $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$, SE: $\frac{d^2 u}{dr^2} - \frac{2\mu\varepsilon}{\hbar^2} u = \left[-\frac{2\mu}{\hbar^2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{l(l+1)}{r^2} \right] u$ with $u(0) = 0$

Define:

$$\kappa = \sqrt{\frac{2\mu\varepsilon}{\hbar^2}}; \quad \rho_0 = \frac{\mu Ze^2}{2\pi\epsilon_0 \kappa \hbar^2} = \frac{Ze^2}{2\pi\epsilon_0 \hbar} \sqrt{\frac{\mu}{2\varepsilon}}; \text{ and let } \rho = \kappa r (\text{dimensionless})$$

$$\frac{d^2 u}{d\rho^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2} \right] u, \quad \text{with } u(0) = 0$$

$$\begin{array}{l|l} \rho \rightarrow 0, \quad \frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u = 0 & \rho \rightarrow \infty, \quad \frac{d^2 u}{d\rho^2} = u \\ \implies u \propto \rho^{l+1} (\text{good}) \text{ or } \rho^{-l} (\text{bad}) & \implies u \propto e^{-\rho} (\text{good}) \text{ or } e^{\rho} (\text{bad}) \end{array}$$

$$\therefore u(\rho) = \rho^{l+1} e^{-\rho} v(\rho)$$

$v(\rho)$ must be s. t. $e^{-\rho} v(\rho)$ decays faster than the rise of ρ^{l+1} as $\rho \rightarrow \infty$

states of definite ang momentum with quantum nos. $l, m : \psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

radial SE: $\frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$

consider free (ionized) electron to be of zero energy (reference).

bound state problem, $E < 0$; $E = -\varepsilon$ where $\varepsilon > 0$

using $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$, SE: $\frac{d^2 u}{dr^2} - \frac{2\mu\varepsilon}{\hbar^2} u = \left[-\frac{2\mu}{\hbar^2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{l(l+1)}{r^2} \right] u$ with $u(0) = 0$

Define:

$$\kappa = \sqrt{\frac{2\mu\varepsilon}{\hbar^2}}; \quad \rho_0 = \frac{\mu Ze^2}{2\pi\epsilon_0 \kappa \hbar^2} = \frac{Ze^2}{2\pi\epsilon_0 \hbar} \sqrt{\frac{\mu}{2\varepsilon}}; \text{ and let } \rho = \kappa r (\text{dimensionless})$$

$$\frac{d^2 u}{d\rho^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2} \right] u, \quad \text{with } u(0) = 0$$

$$\begin{array}{l|l} \rho \rightarrow 0, \quad \frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u = 0 & \rho \rightarrow \infty, \quad \frac{d^2 u}{d\rho^2} = u \\ \implies u \propto \rho^{l+1} (\text{good}) \text{ or } \rho^{-l} (\text{bad}) & \implies u \propto e^{-\rho} (\text{good}) \text{ or } e^{\rho} (\text{bad}) \end{array}$$
$$\therefore u(\rho) = \rho^{l+1} e^{-\rho} v(\rho)$$

$v(\rho)$ must be s. t. $e^{-\rho} v(\rho)$ decays faster than the rise of ρ^{l+1} as $\rho \rightarrow \infty$
and is finite for $\rho \rightarrow 0$

states of definite ang momentum with quantum nos. $l, m : \psi(\vec{r}) = \frac{1}{r} u(r) Y_l^m(\theta, \phi)$

$$\text{radial SE: } \frac{d^2 u}{dr^2} + \frac{2\mu E}{\hbar^2} u = \left[\frac{2\mu V}{\hbar^2} + \frac{l(l+1)}{r^2} \right] u$$

consider free (ionized) electron to be of zero energy (reference).

bound state problem, $E < 0$; $E = -\varepsilon$ where $\varepsilon > 0$

$$\text{using } V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}, \text{ SE: } \frac{d^2 u}{dr^2} - \frac{2\mu\varepsilon}{\hbar^2} u = \left[-\frac{2\mu}{\hbar^2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{l(l+1)}{r^2} \right] u \text{ with } u(0) = 0$$

Define:

$$\kappa = \sqrt{\frac{2\mu\varepsilon}{\hbar^2}}; \quad \rho_0 = \frac{\mu Ze^2}{2\pi\epsilon_0 \kappa \hbar^2} = \frac{Ze^2}{2\pi\epsilon_0 \hbar} \sqrt{\frac{\mu}{2\varepsilon}}; \text{ and let } \rho = \kappa r (\text{dimensionless})$$

$$\frac{d^2 u}{d\rho^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2} \right] u, \quad \text{with } u(0) = 0$$

$$\begin{array}{l|l} \rho \rightarrow 0, \quad \frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u = 0 & \rho \rightarrow \infty, \quad \frac{d^2 u}{d\rho^2} = u \\ \implies u \propto \rho^{l+1} (\text{good}) \text{ or } \rho^{-l} (\text{bad}) & \implies u \propto e^{-\rho} (\text{good}) \text{ or } e^{\rho} (\text{bad}) \end{array}$$

$$\therefore u(\rho) = \rho^{l+1} e^{-\rho} v(\rho)$$

$v(\rho)$ must be s. t. $e^{-\rho} v(\rho)$ decays faster than the rise of ρ^{l+1} as $\rho \rightarrow \infty$
and is finite for $\rho \rightarrow 0$

$$\text{Substituting this in SE: } \rho \frac{d^2 v}{d\rho^2} + [2(l+1) - \rho] \frac{dv}{d\rho} + (\rho_0 - l - 1) v = 0$$

$$\rho \frac{d^2 v}{d\rho^2} + 2(l+1-\rho) \frac{dv}{d\rho} + (\rho_0 - 2(l+1)) v = 0$$

$$\rho \frac{d^2 v}{d\rho^2} + 2(l+1-\rho) \frac{dv}{d\rho} + (\rho_0 - 2(l+1)) v = 0$$

This is a form of equation called Kummer equation,

$$\rho \frac{d^2 v}{d\rho^2} + 2(l+1-\rho) \frac{dv}{d\rho} + (\rho_0 - 2(l+1)) v = 0$$

This is a form of equation called Kummer equation,

Look for power series solution : $v(\rho) = \sum_0^{\infty} c_j \rho^j$

$$\rho \frac{d^2 v}{d\rho^2} + 2(l+1-\rho) \frac{dv}{d\rho} + (\rho_0 - 2(l+1)) v = 0$$

This is a form of equation called Kummer equation,

Look for power series solution : $v(\rho) = \sum_0^{\infty} c_j \rho^j$

demand well-behaved solution everywhere

$$\rho \frac{d^2 v}{d\rho^2} + 2(l+1-\rho) \frac{dv}{d\rho} + (\rho_0 - 2(l+1)) v = 0$$

This is a form of equation called Kummer equation,

Look for power series solution : $v(\rho) = \sum_0^{\infty} c_j \rho^j$

demand well-behaved solution everywhere

This requires, $\rho_0 = 2(l+N)$

$$\rho \frac{d^2 v}{d\rho^2} + 2(l+1-\rho) \frac{dv}{d\rho} + (\rho_0 - 2(l+1)) v = 0$$

This is a form of equation called Kummer equation,

Look for power series solution : $v(\rho) = \sum_0^{\infty} c_j \rho^j$

demand well-behaved solution everywhere

This requires, $\rho_0 = 2(l+N)$

Define : $n = N + l = 1, 2, \dots$ (positive integer; keep in mind : lowest value of l is 0, $\because l \not\leq 0$ integer)

$$\rho \frac{d^2 v}{d\rho^2} + 2(l+1-\rho) \frac{dv}{d\rho} + (\rho_0 - 2(l+1)) v = 0$$

This is a form of equation called Kummer equation,

Look for power series solution : $v(\rho) = \sum_0^{\infty} c_j \rho^j$

demand well-behaved solution everywhere

This requires, $\rho_0 = 2(l+N)$

Define : $n = N + l = 1, 2, \dots$ (positive integer; keep in mind : lowest value of l is 0, $\because l \not\leq 0$ integer)

For given n , $l=0, 1, \dots, n-1$ (min. value of $N = 1$,

$$\rho \frac{d^2 v}{d\rho^2} + 2(l+1-\rho) \frac{dv}{d\rho} + (\rho_0 - 2(l+1)) v = 0$$

This is a form of equation called Kummer equation,

Look for power series solution : $v(\rho) = \sum_0^{\infty} c_j \rho^j$

demand well-behaved solution everywhere

This requires, $\rho_0 = 2(l+N)$

Define : $n = N + l = 1, 2, \dots$ (positive integer; keep in mind : lowest value of l is 0, $\therefore l \not\leq 0$ integer)

For given n , $l=0, 1, \dots, n-1$ (min. value of $N = 1$, $\therefore$ max. value of l is $n-1$)

$$\rho_0 = \frac{\mu Ze^2}{2\pi\epsilon_0\kappa\hbar^2} = \frac{Ze^2}{2\pi\epsilon_0\hbar} \sqrt{\frac{\mu}{2\varepsilon}} = 2n; \quad \therefore E = -\varepsilon = -\frac{(Ze^2)^2}{2(4\pi\epsilon_0)^2\hbar^2} \frac{\mu}{n^2}.$$

$$\rho_0 = \frac{\mu Z e^2}{2\pi\epsilon_0\kappa\hbar^2} = \frac{Z e^2}{2\pi\epsilon_0\hbar} \sqrt{\frac{\mu}{2\varepsilon}} = 2n; \quad \therefore E = -\varepsilon = -\frac{(Z e^2)^2}{2(4\pi\epsilon_0)^2\hbar^2} \frac{\mu}{n^2}.$$

Notice : E has no l dependence, i.e., all states with same value of n but different values of l are degenerate

$$\rho_0 = \frac{\mu Z e^2}{2\pi\epsilon_0\hbar^2} = \frac{Z e^2}{2\pi\epsilon_0\hbar} \sqrt{\frac{\mu}{2\varepsilon}} = 2n; \quad \therefore E = -\varepsilon = -\frac{(Z e^2)^2}{2(4\pi\epsilon_0)^2\hbar^2} \frac{\mu}{n^2}.$$

Notice : E has no l dependence, i.e., all states with same value of n but different values of l are degenerate

This degeneracy is called 'accidental degeneracy'

$$\rho_0 = \frac{\mu Z e^2}{2\pi\epsilon_0\hbar^2} = \frac{Z e^2}{2\pi\epsilon_0\hbar} \sqrt{\frac{\mu}{2\varepsilon}} = 2n; \quad \therefore E = -\varepsilon = -\frac{(Z e^2)^2}{2(4\pi\epsilon_0)^2\hbar^2} \frac{\mu}{n^2}.$$

Notice : E has no l dependence, i.e., all states with same value of n but different values of l are degenerate

This degeneracy is called 'accidental degeneracy'

(For free particle, states with different angular momentum have different energies)

$$\rho_0 = \frac{\mu Ze^2}{2\pi\epsilon_0\hbar^2} = \frac{Ze^2}{2\pi\epsilon_0\hbar} \sqrt{\frac{\mu}{2\varepsilon}} = 2n; \quad \therefore E = -\varepsilon = -\frac{(Ze^2)^2}{2(4\pi\epsilon_0)^2\hbar^2} \frac{\mu}{n^2}.$$

Notice : E has no l dependence, i.e., all states with same value of n but different values of l are degenerate

This degeneracy is called 'accidental degeneracy'

(For free particle, states with different angular momentum have different energies)

The polynomial $v(\rho)$ is a kind of associated Laguerre polynomial of degree $n - l - 1$.

let $a_0 = \frac{\hbar^2}{\mu e^2}$ (Bohr radius, has dimensions of length $\approx 0.0529\text{nm}$).

let $a_0 = \frac{\hbar^2}{\mu e^2}$ (Bohr radius, has dimensions of length $\approx 0.0529\text{nm}$).

write $\frac{1}{4\pi\epsilon_0} \equiv 1$, then, $E_n = -\frac{(Ze^2)^2}{2\hbar^2} \frac{\mu}{n^2} = -\frac{Z^2 e^2}{2a_0} \cdot \frac{1}{n^2}$

let $a_0 = \frac{\hbar^2}{\mu e^2}$ (Bohr radius, has dimensions of length $\approx 0.0529\text{nm}$).

write $\frac{1}{4\pi\epsilon_0} \equiv 1$, then, $E_n = -\frac{(Ze^2)^2}{2\hbar^2} \frac{\mu}{n^2} = -\frac{Z^2 e^2}{2a_0} \cdot \frac{1}{n^2}$

taking $\mu = m_e$,

let $a_0 = \frac{\hbar^2}{\mu e^2}$ (Bohr radius, has dimensions of length $\approx 0.0529\text{nm}$).

write $\frac{1}{4\pi\epsilon_0} \equiv 1$, then, $E_n = -\frac{(Ze^2)^2}{2\hbar^2} \frac{\mu}{n^2} = -\frac{Z^2 e^2}{2a_0} \cdot \frac{1}{n^2}$

taking $\mu = m_e$, $E_n = -\frac{Z^2}{n^2} \times 13.59 \text{ eV} = -\frac{Z^2}{n^2} \times \frac{1}{2} \text{ Hartree}$

let $a_0 = \frac{\hbar^2}{\mu e^2}$ (Bohr radius, has dimensions of length $\approx 0.0529\text{nm}$).

write $\frac{1}{4\pi\epsilon_0} \equiv 1$, then, $E_n = -\frac{(Ze^2)^2}{2\hbar^2} \frac{\mu}{n^2} = -\frac{Z^2 e^2}{2a_0} \cdot \frac{1}{n^2}$

taking $\mu = m_e$, $E_n = -\frac{Z^2}{n^2} \times 13.59 \text{ eV} = -\frac{Z^2}{n^2} \times \frac{1}{2} \text{ Hartree}$

$\rho \equiv \rho_n = \frac{r}{a_0} \frac{Z}{n}$

let $a_0 = \frac{\hbar^2}{\mu e^2}$ (Bohr radius, has dimensions of length $\approx 0.0529\text{nm}$).

write $\frac{1}{4\pi\epsilon_0} \equiv 1$, then, $E_n = -\frac{(Ze^2)^2}{2\hbar^2} \frac{\mu}{n^2} = -\frac{Z^2 e^2}{2a_0} \cdot \frac{1}{n^2}$

taking $\mu = m_e$, $E_n = -\frac{Z^2}{n^2} \times 13.59 \text{ eV} = -\frac{Z^2}{n^2} \times \frac{1}{2} \text{ Hartree}$

$\rho \equiv \rho_n = \frac{r}{a_0} \frac{Z}{n}$

The complete solution $\psi_{n,l,m} = \mathcal{R}_{n,l}(r) Y_l^m(\Omega)$

let $a_0 = \frac{\hbar^2}{\mu e^2}$ (Bohr radius, has dimensions of length $\approx 0.0529\text{nm}$).

write $\frac{1}{4\pi\epsilon_0} \equiv 1$, then, $E_n = -\frac{(Ze^2)^2}{2\hbar^2} \frac{\mu}{n^2} = -\frac{Z^2 e^2}{2a_0} \cdot \frac{1}{n^2}$

taking $\mu = m_e$, $E_n = -\frac{Z^2}{n^2} \times 13.59 \text{ eV} = -\frac{Z^2}{n^2} \times \frac{1}{2} \text{ Hartree}$

$\rho \equiv \rho_n = \frac{r}{a_0} \frac{Z}{n}$

The complete solution $\psi_{n,l,m} = \mathcal{R}_{n,l}(r) Y_l^m(\Omega)$

$$\mathcal{R}_{n,l}(r) = \text{const.} r^l \cdot e^{-\frac{rZ}{na_0}} \cdot v(r)$$

let $a_0 = \frac{\hbar^2}{\mu e^2}$ (Bohr radius, has dimensions of length $\approx 0.0529\text{nm}$).

write $\frac{1}{4\pi\epsilon_0} \equiv 1$, then, $E_n = -\frac{(Ze^2)^2}{2\hbar^2} \frac{\mu}{n^2} = -\frac{Z^2 e^2}{2a_0} \cdot \frac{1}{n^2}$

taking $\mu = m_e$, $E_n = -\frac{Z^2}{n^2} \times 13.59 \text{ eV} = -\frac{Z^2}{n^2} \times \frac{1}{2} \text{ Hartree}$

$\rho \equiv \rho_n = \frac{r}{a_0} \frac{Z}{n}$

The complete solution $\psi_{n,l,m} = \mathcal{R}_{n,l}(r) Y_l^m(\Omega)$

$$\mathcal{R}_{n,l}(r) = \text{const.} r^l \cdot e^{-\frac{rZ}{na_0}} \cdot v(r)$$

where $v(r)$ is a Laguerre polynomial

Orthonormality $\implies \langle \psi_{n', l', m'} | \psi_{n, l, m} \rangle = \delta_{n' n} \delta_{l' l} \delta_{m' m}.$

Orthonormality $\implies \langle \psi_{n', l', m'} | \psi_{n, l, m} \rangle = \delta_{n' n} \delta_{l' l} \delta_{m' m}.$

We already know that $\int Y_{l'}^{m'}(\Omega) Y_l^m(\Omega) d\Omega = \delta_{l' l} \delta_{m' m}$

Orthonormality $\implies \langle \psi_{n', l', m'} | \psi_{n, l, m} \rangle = \delta_{n' n} \delta_{l' l} \delta_{m' m}$.

We already know that $\int Y_{l'}^{m'}(\Omega) Y_l^m(\Omega) d\Omega = \delta_{l' l} \delta_{m' m}$

$$\therefore \int \mathcal{R}_{n', l}(r) \mathcal{R}_{n, l}(r) r^2 dr = \delta_{n' n}$$

Orthonormality $\implies \langle \psi_{n', l', m'} | \psi_{n, l, m} \rangle = \delta_{n'n} \delta_{l'l} \delta_{m'm}$.

We already know that $\int Y_{l'}^{m'}(\Omega) Y_l^m(\Omega) d\Omega = \delta_{l'l} \delta_{m'm}$

$$\therefore \int \mathcal{R}_{n', l}(r) \mathcal{R}_{n, l}(r) r^2 dr = \delta_{n'n}$$

$$\mathcal{R}_{1,0}(r) = 2 \left(\frac{Z}{a_0} \right)^{\frac{3}{2}} e^{-\frac{Zr}{a_0}}$$

Orthonormality $\implies \langle \psi_{n', l', m'} | \psi_{n, l, m} \rangle = \delta_{n' n} \delta_{l' l} \delta_{m' m}$.

We already know that $\int Y_{l'}^{m'}(\Omega) Y_l^m(\Omega) d\Omega = \delta_{l' l} \delta_{m' m}$

$$\therefore \int \mathcal{R}_{n', l}(r) \mathcal{R}_{n, l}(r) r^2 dr = \delta_{n' n}$$

$$\mathcal{R}_{1, 0}(r) = 2 \left(\frac{Z}{a_0} \right)^{\frac{3}{2}} e^{-\frac{Zr}{a_0}}$$

$$\mathcal{R}_{2, 0}(r) = \left(\frac{Z}{2a_0} \right)^{\frac{3}{2}} \left(2 - \frac{Zr}{a_0} \right) e^{-\frac{Zr}{2a_0}}$$

Orthonormality $\implies \langle \psi_{n', l', m'} | \psi_{n, l, m} \rangle = \delta_{n' n} \delta_{l' l} \delta_{m' m}$.

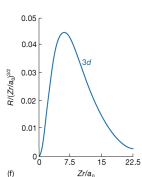
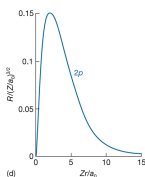
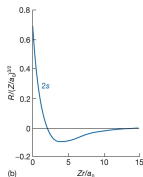
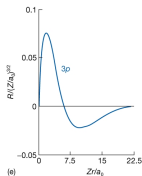
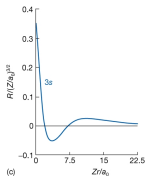
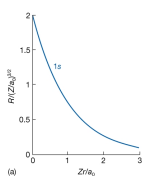
We already know that $\int Y_{l', m'}(\Omega) Y_{l, m}(\Omega) d\Omega = \delta_{l' l} \delta_{m' m}$

$\therefore \int \mathcal{R}_{n', l}(r) \mathcal{R}_{n, l}(r) r^2 dr = \delta_{n' n}$

$$\mathcal{R}_{1,0}(r) = 2 \left(\frac{Z}{a_0} \right)^{\frac{3}{2}} e^{-\frac{Zr}{a_0}}$$

$$\mathcal{R}_{2,0}(r) = \left(\frac{Z}{2a_0} \right)^{\frac{3}{2}} \left(2 - \frac{Zr}{a_0} \right) e^{-\frac{Zr}{2a_0}}$$

$$\mathcal{R}_{2,1}(r) = \frac{1}{\sqrt{3}} \left(\frac{Z}{2a_0} \right)^{\frac{3}{2}} \left(\frac{Zr}{a_0} \right) e^{-\frac{Zr}{2a_0}}$$



Orthonormality $\implies \langle \psi_{n', l', m'} | \psi_{n, l, m} \rangle = \delta_{n' n} \delta_{l' l} \delta_{m' m}$.

We already know that $\int Y_{l'}^{m'}(\Omega) Y_l^m(\Omega) d\Omega = \delta_{l' l} \delta_{m' m}$

$\therefore \int \mathcal{R}_{n', l}(r) \mathcal{R}_{n, l}(r) r^2 dr = \delta_{n' n}$

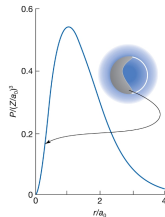
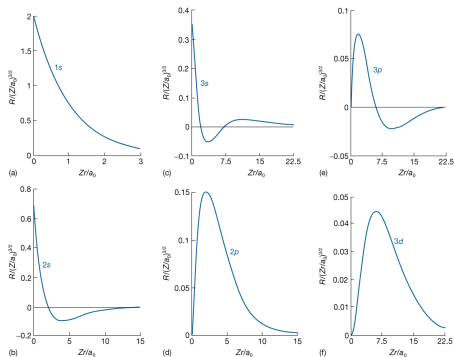
$$\mathcal{R}_{1,0}(r) = 2 \left(\frac{Z}{a_0} \right)^{\frac{3}{2}} e^{-\frac{Zr}{a_0}}$$

$$\mathcal{R}_{2,0}(r) = \left(\frac{Z}{2a_0} \right)^{\frac{3}{2}} \left(2 - \frac{Zr}{a_0} \right) e^{-\frac{Zr}{2a_0}}$$

$$\mathcal{R}_{2,1}(r) = \frac{1}{\sqrt{3}} \left(\frac{Z}{2a_0} \right)^{\frac{3}{2}} \left(\frac{Zr}{a_0} \right) e^{-\frac{Zr}{2a_0}}$$

radial probability distribution
for 1s orbital

$$P(r)dr = 4\pi r^2 dr \times (\mathcal{R}_{1,0})^2$$

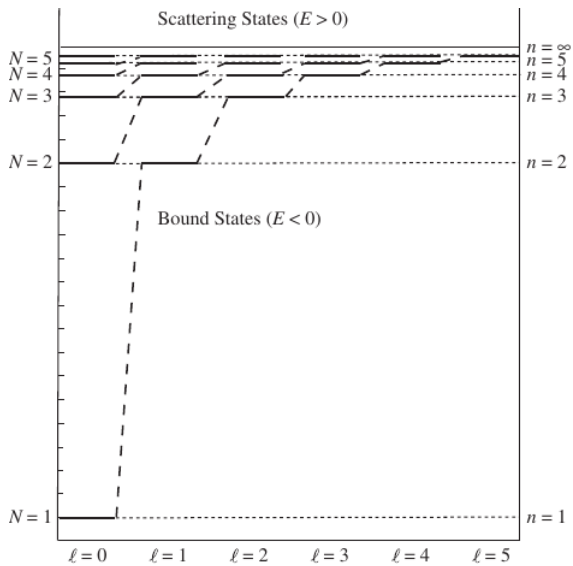


Useful integrals:

$$\begin{aligned}\int \mathcal{R}_{n,l}^2 r^2 \left(\frac{Zr}{a_0}\right)^\nu dr &= \frac{1}{2} n^2 [5n^2 + 1 - 3l(l+1)], \text{ for } \nu = 2 \\ &= \frac{1}{2} [3n^2 - l(l+1)], \text{ for } \nu = 1 \\ &= \frac{1}{n^2}, \text{ for } \nu = -1 \\ &= \frac{2}{n^3(2l+1)}, \text{ for } \nu = -2\end{aligned}$$

Using this, it can be shown that

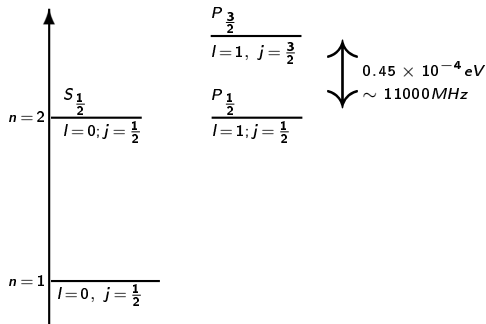
$$\begin{aligned}\left\langle \frac{1}{r} \right\rangle &= \langle n, l, m | \frac{1}{r} | n, l, m \rangle = \int r^2 dr \cdot \psi_{n,l,m}^*(r, \theta, \phi) \psi_{n,l,m}(r, \theta, \phi) d\Omega \\ &= \langle \mathcal{R}_{n,l} | \frac{1}{r} | \mathcal{R}_{n,l} \rangle \langle Y_l^m | Y_l^m \rangle = \int r^2 \cdot \mathcal{R}_{n,l}^2(r) \cdot \frac{1}{r} dr = \frac{Z}{a_0 n^2}\end{aligned}$$



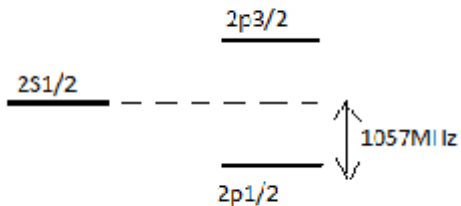
relativistic and spin corrections (first order shift depends only on j):

$$\therefore E_{n,l,j=l\pm\frac{1}{2}} = E_n^{(0)} - \frac{(Z\alpha)^4}{2n^4} \mu c^2 \left[\frac{n}{j+\frac{1}{2}} - \frac{3}{4} \right], \quad \alpha = \frac{e^2}{\hbar c} \approx \frac{1}{137} : \text{fine structure constant.}$$

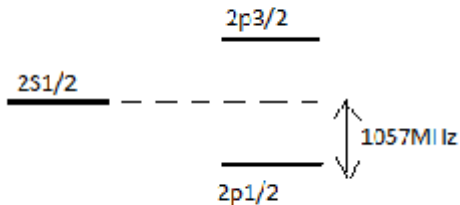
energy diagram for $Z = 1$ (H) looks like the following:



The reality is a little different for $l=1$ states

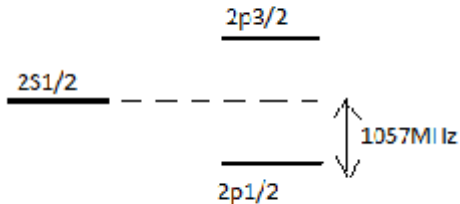


The reality is a little different for $l=1$ states



Gap between $2S_{\frac{1}{2}}$ and $2P_{\frac{1}{2}}$ is 1057 MHz.

The reality is a little different for $l=1$ states



Gap between $2S_{\frac{1}{2}}$ and $2P_{\frac{1}{2}}$ is 1057 MHz.

Lamb shift : $2S_{\frac{1}{2}}$ is raised by 1040 MHz and $2P_{\frac{1}{2}}$ is lowered by 17MHz

Multielectronic atoms

Multielectronic atoms : interelectronic repulsion

Multielectronic atoms : interelectronic repulsion
an empirical solution : shielded nuclear charge

Multielectronic atoms : interelectronic repulsion
an empirical solution : shielded nuclear charge

$$E_n = -\frac{Z^{*2}}{n^2} \times 13.59 \text{ eV}$$

Multielectronic atoms : interelectronic repulsion
an empirical solution : shielded nuclear charge

$$E_n = -\frac{Z^{*2}}{n^2} \times 13.59 \text{ eV}$$

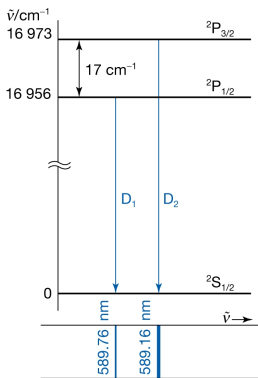
spin-orbit coupling (addition of angular momenta)

additional term in Hamiltonian, $A\vec{L}\cdot\vec{S}$; $j = |l - s|, |l - s| + 1, \dots, l + s$

Sodium D lines: simple

explanation:

single electron:



spin-orbit coupling (addition of angular momenta)

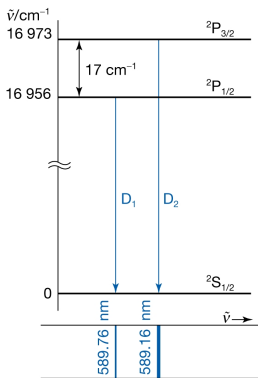
additional term in Hamiltonian, $A\vec{L}\cdot\vec{S}$; $j = |l - s|, |l - s| + 1, \dots, l + s$

Sodium D lines: simple

explanation:

single electron:

ground state: $l = 0, s = \frac{1}{2}, j = \frac{1}{2}$



spin-orbit coupling (addition of angular momenta)

additional term in Hamiltonian, $\hat{A}\vec{l}\cdot\vec{s}$; $j = |l - s|, |l - s| + 1, \dots, l + s$

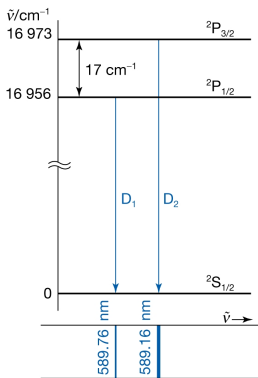
Sodium D lines: simple

explanation:

single electron:

ground state: $l = 0, s = \frac{1}{2}, j = \frac{1}{2}$

excited state: $l = 1, s = \frac{1}{2}, j = \frac{1}{2}, \frac{3}{2}$



spin-orbit coupling (addition of angular momenta)

additional term in Hamiltonian, $\widehat{\vec{L}} \cdot \widehat{\vec{S}}$; $j = |l - s|, |l - s| + 1, \dots, l + s$

Sodium D lines: simple

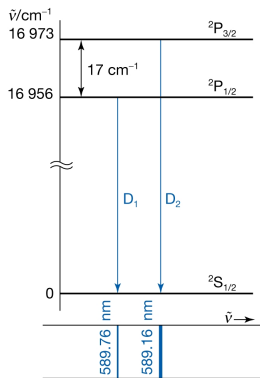
explanation:

single electron:

ground state: $l = 0, s = \frac{1}{2}, j = \frac{1}{2}$

excited state: $l = 1, s = \frac{1}{2}, j = \frac{1}{2}, \frac{3}{2}$

$$\langle j \ l \ s | \widehat{\vec{L}} \cdot \widehat{\vec{S}} | j \ l \ s \rangle =$$



spin-orbit coupling (addition of angular momenta)

additional term in Hamiltonian, $A\hat{L}\cdot\hat{S}$; $j = |l - s|, |l - s| + 1, \dots, l + s$

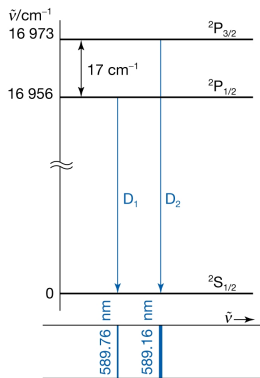
Sodium D lines: simple

explanation:

single electron:

ground state: $l = 0, s = \frac{1}{2}, j = \frac{1}{2}$

excited state: $l = 1, s = \frac{1}{2}, j = \frac{1}{2}, \frac{3}{2}$



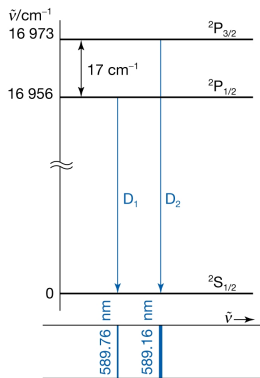
$$\begin{aligned} \langle j \mid s \mid \hat{L} \cdot \hat{S} \mid j \mid s \rangle &= \\ &= \frac{1}{2} \langle j \mid s \mid \hat{j}^2 - \hat{l}^2 - \hat{s}^2 \mid j \mid s \rangle \end{aligned}$$

spin-orbit coupling (addition of angular momenta)

additional term in Hamiltonian, $A\hat{L}\cdot\hat{S}$; $j = |l - s|, |l - s| + 1, \dots, l + s$

Sodium D lines: simple

explanation:



single electron:

ground state: $l = 0, s = \frac{1}{2}, j = \frac{1}{2}$

excited state: $l = 1, s = \frac{1}{2}, j = \frac{1}{2}, \frac{3}{2}$

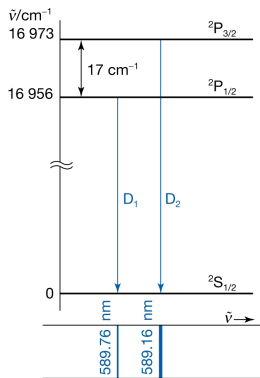
$$\begin{aligned}
 \langle j \ l \ s | \hat{L} \cdot \hat{S} | j \ l \ s \rangle &= \\
 &= \frac{1}{2} \langle j \ l \ s | \hat{j}^2 - \hat{l}^2 - \hat{s}^2 | j \ l \ s \rangle \\
 &= \frac{\hbar^2}{2} [j(j+1) - l(l+1) - s(s+1)]
 \end{aligned}$$

spin-orbit coupling (addition of angular momenta)

additional term in Hamiltonian, $\widehat{\vec{L}} \cdot \widehat{\vec{S}}$; $j = |l - s|, |l - s| + 1, \dots, l + s$

Sodium D lines: simple

explanation:



single electron:

ground state: $l = 0, s = \frac{1}{2}, j = \frac{1}{2}$

excited state: $l = 1, s = \frac{1}{2}, j = \frac{1}{2}, \frac{3}{2}$

$$\begin{aligned} \langle j \ l \ s | \widehat{\vec{L}} \cdot \widehat{\vec{S}} | j \ l \ s \rangle &= \\ &= \frac{1}{2} \langle j \ l \ s | \widehat{j}^2 - \widehat{l}^2 - \widehat{s}^2 | j \ l \ s \rangle \\ &= \frac{\hbar^2}{2} [j(j+1) - l(l+1) - s(s+1)] \end{aligned}$$

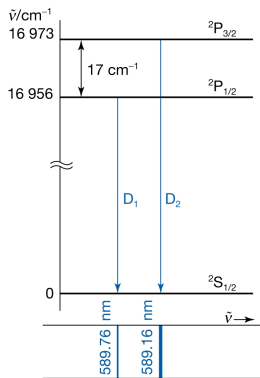
For $j = \frac{3}{2}$, $\Delta E_{\frac{3}{2}} = \frac{1}{2} \hbar c A$

spin-orbit coupling (addition of angular momenta)

additional term in Hamiltonian, $\widehat{\vec{L}} \cdot \widehat{\vec{S}}$; $j = |l - s|, |l - s| + 1, \dots, l + s$

Sodium D lines: simple

explanation:



single electron:

ground state: $l = 0, s = \frac{1}{2}, j = \frac{1}{2}$

excited state: $l = 1, s = \frac{1}{2}, j = \frac{1}{2}, \frac{3}{2}$

$$\begin{aligned} \langle j \ l \ s | \widehat{\vec{L}} \cdot \widehat{\vec{S}} | j \ l \ s \rangle &= \\ &= \frac{1}{2} \langle j \ l \ s | \widehat{j}^2 - \widehat{l}^2 - \widehat{s}^2 | j \ l \ s \rangle \\ &= \frac{\hbar^2}{2} [j(j+1) - l(l+1) - s(s+1)] \end{aligned}$$

For $j = \frac{3}{2}, \Delta E_{\frac{3}{2}} = \frac{1}{2} \hbar c A$

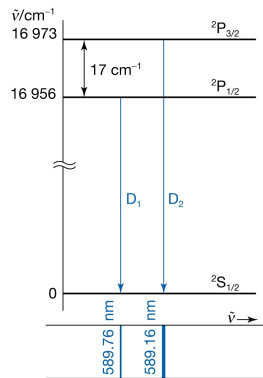
For $j = \frac{1}{2}, \Delta E_{\frac{1}{2}} = -\hbar c A$

spin-orbit coupling (addition of angular momenta)

additional term in Hamiltonian, $\widehat{A}\vec{l}\cdot\vec{s}$; $j = |l - s|, |l - s| + 1, \dots, l + s$

Sodium D lines: simple

explanation:



single electron:

ground state: $l = 0, s = \frac{1}{2}, j = \frac{1}{2}$

excited state: $l = 1, s = \frac{1}{2}, j = \frac{1}{2}, \frac{3}{2}$

$$\begin{aligned} \langle j \, l \, s | \widehat{l} \cdot \widehat{s} | j \, l \, s \rangle &= \\ &= \frac{1}{2} \langle j \, l \, s | \widehat{j}^2 - \widehat{l}^2 - \widehat{s}^2 | j \, l \, s \rangle \\ &= \frac{\hbar^2}{2} [j(j+1) - l(l+1) - s(s+1)] \end{aligned}$$

$$\text{For } j = \frac{3}{2}, \Delta E_{\frac{3}{2}} = \frac{1}{2} \hbar c A$$

$$\text{For } j = \frac{1}{2}, \Delta E_{\frac{1}{2}} = -\hbar c A$$

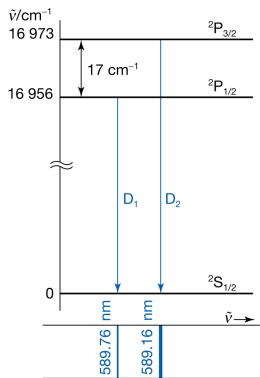
$$\Delta \bar{\nu} = \frac{3}{2} A = 17.2 \text{ cm}^{-1}$$

spin-orbit coupling (addition of angular momenta)

additional term in Hamiltonian, $\widehat{A}\vec{l}\cdot\vec{s}$; $j = |l - s|, |l - s| + 1, \dots, l + s$

Sodium D lines: simple

explanation:



single electron:

ground state: $l = 0, s = \frac{1}{2}, j = \frac{1}{2}$

excited state: $l = 1, s = \frac{1}{2}, j = \frac{1}{2}, \frac{3}{2}$

$$\begin{aligned} \langle j | s | \widehat{l} \cdot \vec{s} | j | s \rangle &= \\ &= \frac{1}{2} \langle j | s | \widehat{j}^2 - \widehat{l}^2 - \widehat{s}^2 | j | s \rangle \\ &= \frac{\hbar^2}{2} [j(j+1) - l(l+1) - s(s+1)] \end{aligned}$$

$$\text{For } j = \frac{3}{2}, \Delta E_{\frac{3}{2}} = \frac{1}{2} \hbar c A$$

$$\text{For } j = \frac{1}{2}, \Delta E_{\frac{1}{2}} = -\hbar c A$$

$$\Delta \bar{\nu} = \frac{3}{2} A = 17.2 \text{ cm}^{-1}$$

$$\therefore A = 11.5 \text{ cm}^{-1}$$